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The Editors
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Dear Editors,

We submit for your consideration the manuscript “*Structure-Aware Row Scaling for Block-Structured Mixed-Integer Programs*” as a Research Article.

Model generators build mixed-integer programs (MIPs) component by component and know, before export, which rows are local to a component, which variables it owns, and which rows couple components—structure that the flat coefficient matrix handed to the solver discards. The paper asks whether this pre-export metadata can be reused as numerical information to drive positive row scaling, a transformation that changes the numbers the solver sees while leaving the feasible set, objective, and integer restrictions exactly unchanged. We believe the work fits IJNME’s scope in numerical methods for engineering computation: it concerns the numerical representation of generated engineering optimization models and the interaction between that representation and commercial solvers, evaluated on a hydrothermal-dispatch platform used as a demanding block-structured testbed rather than as a domain application.

The central contribution is methodological as much as algorithmic. Across three instance classes, two commercial solvers, and two optimality tolerances, we separate four kinds of evidence that are easily conflated—algebraic equivalence, numerical representation, solver-facing conditioning, and matched per-solver runtime—and show that they need not agree. Positive row scaling is exact; structure-aware scaling robustly improves solver-facing numerical diagnostics; and runtime improves under Gurobi but not robustly under CPLEX. A role-blind control then isolates the runtime lever as a cheap per-row normalization rather than the row-role metadata, with a class-dependent outcome that we trace to a coverage gap over the highest-magnitude rows. The result is a clean, controlled demonstration that the rule which best conditions a matrix is not the one that solves fastest—a cautionary and, we hope, useful lesson for numerical preprocessing of generated MIPs. We report the negative and mixed results (the CPLEX neutrality, and the class in which the structure-blind control loses) as fully as the positive ones.

The work is original and is not under consideration elsewhere; no part has been previously published. The manuscript has a single author, who is responsible for all of its content. A reproducibility package—cleaned source code, the redistributable dispatch and synthetic instances, aggregate per-scenario results, and the scripts that rebuild every table and figure—is publicly available and archived with a DOI, as stated in the Data Availability Statement; the experiments were carried out on the ClusterUY high-performance computing platform. The manuscript declares no conflicts of interest and received no specific funding.

We would be glad to suggest suitable reviewers on request, and we thank you for considering our submission.

Sincerely,

Mathias Rodríguez Castro